Linear Algebra (MT-1004)

Homework#13

Sections: B,C & D

Instructor: Muhammad Ali

- Find the standard matrix corresponding to the following linear transformation with respect to the given basis:

$$T : \mathbb{R}^2 \to \mathbb{R}^3$$

$$T \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a+2b \\ -a \\ b \end{bmatrix}$$

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \end{bmatrix} \right\}, \mathcal{C} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$